965349

Course Name: **Database Management System**

Course Outcome

CO1: Understand the concept of database management systems and Relational database.
CO2: Identify the various data model used in database design.
CO3: Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
CO4: Create and populate a RDBMS for a real-life application, with constraints and keys using SQL.
CO5: Select the information from a database by formulating complex queries in SQL.
CO6: Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
CO7: Discuss indexing mechanisms for efficient retrieval of information from a database.
CO8: Discuss recovery system and be familiar with introduction to web database, distributed databases.

Printed Pages: 4

University Roll No. .....

End Term Examination, Odd Semester 2024-25
B.Tech. (CSE), IInd Year, IIIrd Semester
BCSC 1003 - Database Management System

Time: 3 Hours

Maximum Marks: 50

Section – A

Attempt All Questions

4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Consider a library management system. Discuss why a DBMS is a better choice for implementation compared to a file system.	4	1	R	F
2	Explain the following basic operations of relational algebra with examples: • Selection (σ) • Projection (π) • Union (∪) • Set Difference (−)	4	3	U	C
3	a. Discuss the term data independency and its types. b. Explain DDL, DML and DCL with example.	4	1	R	C
4	A Relational schema R has four attributes R(W,X,Y,Z) with Functional Dependencies :	4	4	E	P

	{W→X, X→Y, Y→Z, Z→X}. Schema R is decomposed into R1(W,X), R2(X,Y) & R3(X,Z). Check in this decomposition that dependency is preserved or not.				
5	Define deadlock in the context of DBMS. Explain the difference between deadlock prevention, deadlock avoidance, and deadlock detection. Provide examples of each. OR Consider a database for student records containing fields like Student ID, Name, Address, and Phone Number. However, it's observed that some students share the same address. Explain the concept of normalization and identify which normalization form would be most suitable to address this redundancy.	4	6	An	C

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Discuss the term Serializability? Differentiate between Conflict and View Serializability by using an example?	3	5	E	P
7	Briefly discuss data fragmentation in DBMS? Why is it used in distributed databases?	3	8	U	C
8	Consider a relation schema R = (A, B, C, D, E, F) on which the following functional dependencies hold: {A→ B, BC→D, E→ C, D → A} Identify the Candidate Key(s) of R.	3	6	A	C
9	Explain the anomalies that can occur in an unnormalized database (insertion, deletion, and update anomalies) with examples.	3	6	U	F
10	A customer places an order on e-commerce platform, which involves multiple database operations (update inventory, create order, process payment). Explain how the transaction properties (atomicity, consistency, isolation, durability) ensures data	3	8	U	C

	integrity. Also illustrate the complete transaction state diagram for this scenario. OR Define the following terms with examples: • Candidate Key • Primary Key • Partial Dependency • Transitive Dependency				

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Briefly discuss on the Two-phase locking protocol(2PL) and Time stamping protocol used in concurrency control. How does it guarantee serializability?	5	8	R	F
12	Differentiate between dense and sparse indexing. Provide examples of each. Under what conditions would sparse indexing be preferred over dense indexing?	5	7	A	C
13	Consider the following schedule involving 2 transactions: S: R1(x);R2(x);R2(y);W2(y);R1(y);W1(x) Check whether schedule S is Conflict and View Serializable or not? OR An online library system stores its database across three branches: • Branch 1 stores information about books and their availability. • Branch 2 stores information about members.	5	8	An	C

	• Branch 3 stores transaction details like book borrow and return dates. **Questions:** 1. A user wants to borrow a book. Explain how the distributed database system would handle the query to ensure the book is available and update the transaction details. 2. What challenges might arise if one branch becomes temporarily unavailable? How can replication help in this scenario?				